PDKM BioTech
Chemicals & Medicine Div. of KM Group

Signature Manifestation

Batch Record Approval Signatures :

Prepared By: Arya Stark Date: 20-Aug-2023
Arya Stark
Associate Manager, Quality Assurance
MKPD Bio Tech

Approved By: Lannister Twin Date: 21-Aug-2023
Tyrion Lannister
Manager, Drug Manufacturing and Operations
MKPD Bio Tech

Clegane Sandor Date: 22-Aug-2023
Sandor Clegane
Lead Specialist - Quality Assurance
MKPD Bio Tech

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

1. Bill Of Material

Serial Number	Material ID	Material Description	Lot Number	Quantity	Expiry Date	Issued By	Received By
1	A2R7F9	Acetaminophen	ST8910	100ML	31-JUL-2024		
2	K8J3M5	Ascorbic Acid	ST5402	1G	01-AUG-2024		
3	B6N2P4	Magnesium Stearate	ST8828	22.5L	09-SEP-2025		
4	Q1W5E7	Microcrystalline Cellulose	ST3899	4L	10-NOV-2024		
5	X9C4V6	Ethanol	ST2600	12G	01-JUN-2025		
6	G3T8H2	Sodium Chloride	ST7737	1G	04-JUN-2025		
7	Z7Y6U1	Lactose Monohydrate	ST1297	3ML	19-SEP-2025		
8	D5F1R9	Polyethylene Glycol	ST8552	1G	25-SEP-2025		
9	L2K6J4	Propylene Glycol	ST4355	100ML	12-DEC-2025		
10	M4N9B8	Hydroxypropyl Methylcellulose	ST7516	1ML	17-FEB-2026	20-Aug-23	20-Aug-23
11	E7Q1W5	Croscarmellose Sodium	ST9767	2L	11-JAN-2025		
12	V6X2C3	Povidone	ST1162	1G	04-DEC2024		
13	H9G4T7	Sodium Lauryl Sulfate	ST5661	10ML	10-NOV-2024		
14	U1Z8Y5	Dimethyl Sulfoxide (DMSO)	ST6518	1ML	12-OCT-2024		
15	P3D6F2	Gelatin	ST2287	1G	21-JAN-2024		
16	R5A7E9	Polyvinyl Alcohol (PVA)	ST7663	21.2L	23-AUG-2024		
17	T7H2J6	Stearic Acid	ST3796	100ML	29-SEP-2024		
18	Y4V8K1	Titanium Dioxide	ST2610	5L	04-JUN-2024		

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

2. Equipment List

Serial Number	Equipment	Manufacturer	Equipment Number	Experiemnt Used	Initials
1	High-Performance Liquid Chromatography (HPLC) System	Thermo Fisher Scientific	HPLC-05	✓ Yes No NA Caliberation Due Date : 31-Dec-2025	
2	Gas Chromatography-Mass Spectrometry (GC-MS)	Thermo Fisher Scientific	GCMS-A11	✓ Yes No NA Caliberation Due Date : 31-Dec-2025	
3	Fourier-Transform Infrared (FTIR) Spectrometer	Thermo Fisher Scientific		Yes ✓ No NA Caliberation Due Date :	
4	Ultraviolet-Visible (UV-Vis) Spectrophotometer	Thermo Fisher Scientific	UVVIS-07	✓ Yes No NA Caliberation Due Date : 7-June-2026	
5	Liquid Chromatography-Mass Spectrometry (LC-MS)	Waters Corporation	LSCM-09	✓ Yes No NA Caliberation Due Date : 9-Jan-2028	
6	Differential Scanning Calorimeter (DSC)	Waters Corporation	DSC-05	✓ Yes No NA Caliberation Due Date : 9-Jan-2029	
7	Nuclear Magnetic Resonance (NMR) Spectrometer	PerkinElmer		Yes ✓ No NA Caliberation Due Date :	
8	Particle Size Analyzer	PerkinElmer		Yes ✓ No NA Caliberation Due Date :	
9	Dissolution Tester	PerkinElmer	DT-005	✓ Yes No NA Caliberation Due Date : 11-June-2026	

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

3. Weighing and Dispensing Form

	Recorded By:	Verified By:
1. Weigh Acetaminophen and excipients accurately using Mettler Toledo XS205. 2. Record weights and lot numbers in the batch record.		
Ingredients: - Active Pharmaceutical Ingredient (API): PDKM-007 - Excipients: AL-01-05, AL-01-07		
Equipment Used: WM-03, WM-04		
Weighing Details: - API Weight: 25 mg - Excipient 1 Weight: 3 mg - Excipient 2 Weight: 9 mg	20-Aug-2023	20-Aug-23
Lot Numbers: - API Lot: ST-PDKM-007-0090 - Excipient 1 Lot: WM-03-ST-0712 - Excipient 2 Lot: WM-04-ST-0829		

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

4. Granulation

	Recorded By:	Verified By:
1. Combine weighed materials in Glatt GPCG Pro-5. 2. Add binder solution gradually while mixing. 3. Granulate the mixture to form uniform wet mass.	20-Aug-2023	20-Aug-23
Equipment Used: GR-003		
Binder Solution Details: - Binder Solution Used: BS-05 - Volume Added: 50mL		
Granulation Parameters: - Mixing Time: 59 sec - Granulation Time: 360 sec		

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

5. Drying

	Recorded By:	Verified By:
1. Transfer wet mass to GPCG Pro-5. 2. Dry the granules under controlled temperature (70°C) and airflow conditions.		
Equipment Used: AirDryerBed-096		
Drying Parameters: - Temperature: 73°c - Airflow Rate: 0.56 m3/s - Drying Time: 46 Hrs	20-Aug-23	20 Aug-23

6. Milling

	Recorded By:	Verified By:
1. Mill dried granules using FitzMill D6A to achieve desired particle size.		
Equipment Used: ML-095		
Milling Details: - Milling Time: 15 min - Particle Size Achieved: 0.4 mm3	21-Aug-23	21-Aug-23

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

7. Blending

	Recorded By:	Verified By:
1. Mix milled granules with additional excipients in V Blender. 2. Ensure uniform blend using 15 minutes of blending.		
Equipment Used: BL-0083		
Blending Details: - Blending Time: 30 mins - Excipient Homogeneity: 3:1	21-Aug-2023	21-Aug-23

8. Compression

	Recorded By:	Verified By:
1. Load blended mixture into Manesty F3 tablet press 2. Compress tablets at 20 kN compression force and speed of 100 tablets/minute.		
Equipment Used: COMACT-05		
Compression Details: - Compression Force: 0.78 N - Compression Speed: 0.3 m/s - Number of Tablets Compressed: 1024	21-Aug-2023	21-Aug-23

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

9. Coating

	Recorded By:	Verified By:
1. Transfer compressed tablets to Vector Hi-Coater. 2. Apply Opadry coating solution as per formulation requirements. 3. Dry coated tablets in controlled environment (24 hours at 25°C).		
Equipment Used: COA-213		
Coating Details: - Coating Solution Used: PDKM-005-CO - Coating Time: 20 mins - Coating Thickness Achieved: 90 um	21-Aug-23	21 Aug 23